



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO EXPRESS SERVICE STATION.

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

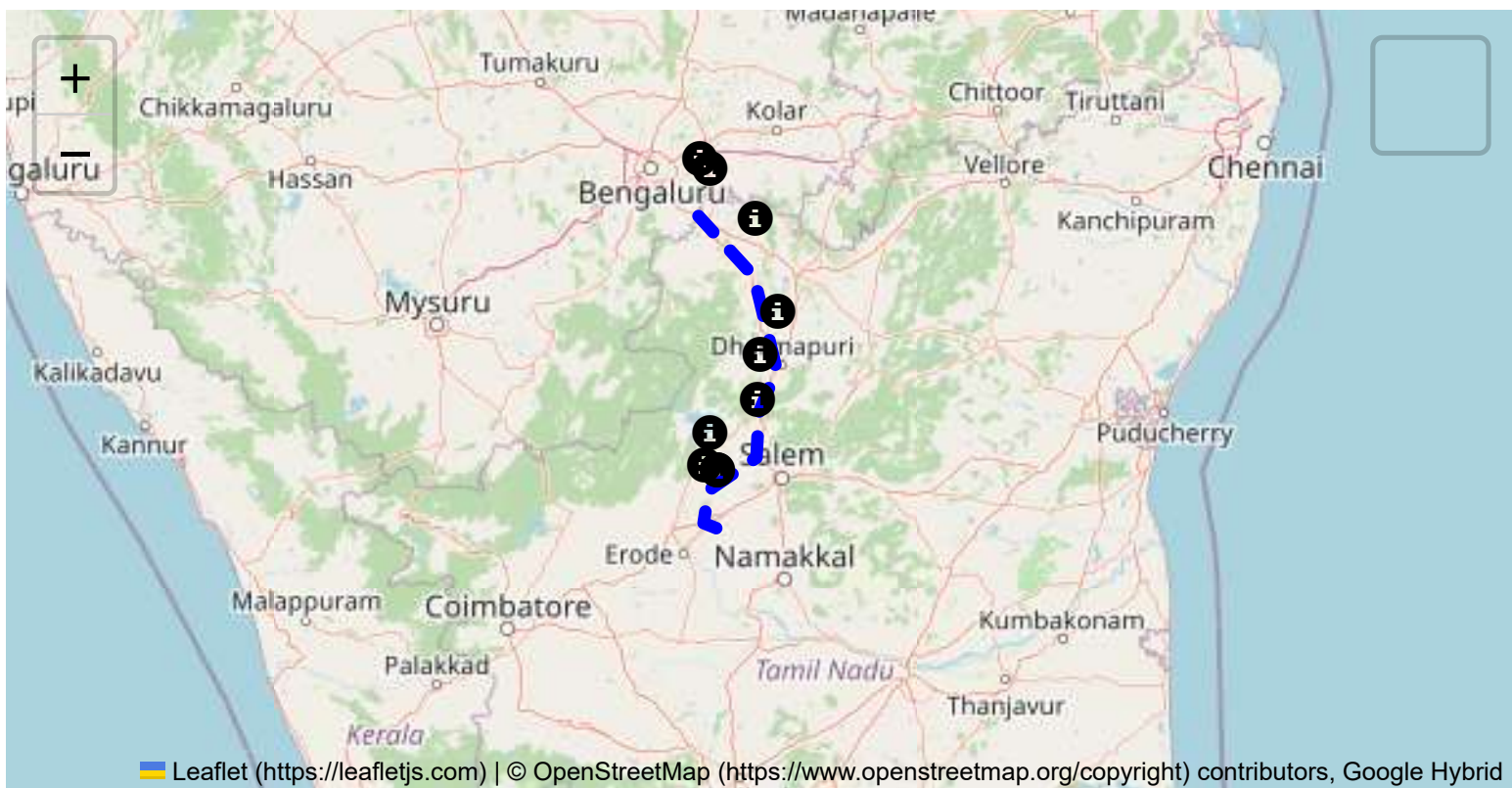
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

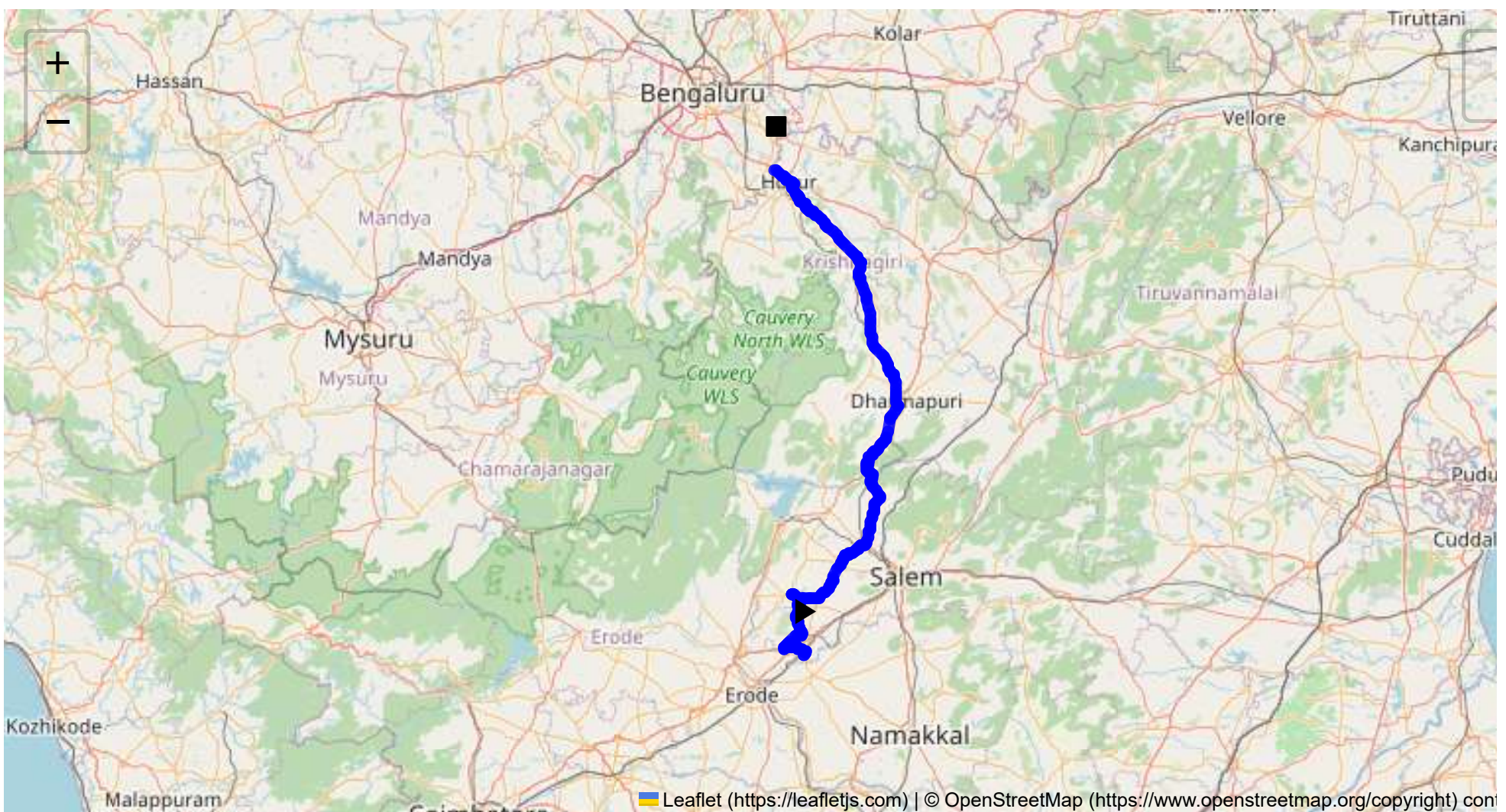
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 200.05 km
Estimated Duration: 4.1 hours
Adjusted Duration (Heavy Vehicle): 5.2 hours
Start: (11.4381, 77.8734)
End: (12.765539, 77.790925)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 200.05 kilometers and passes through several key points, starting at CVQF+23W in Sangagiri and concluding at QQ8R+998 in Zuzuvadi, Hosur. The journey traverses a mix of rural and urban areas, intersecting with several highways and bypasses that serve as major connectors in Tamil Nadu.

Weather Conditions

Tamil Nadu typically experiences a tropical climate with hot summers, moderate monsoons, and mild winters. During the monsoon season (June to September), the region may face heavy rainfall that can lead to waterlogging and reduced visibility. Drivers should be prepared for potential weather-related hazards, such as slippery roads and flash floods, especially near Thoppur and Omalur.

Traffic Patterns

Traffic congestion is common during peak hours (8:00 AM - 10:00 AM and 5:00 PM - 7:00 PM) on major highways, particularly near urban areas like Hosur and Salem. Thoppur, being an industrialized region, can also experience heavy traffic. Travel may be smoother on the rural stretches of the route, depending on local events and agricultural activities.

Road Quality and Infrastructure

The road quality varies along the route. Highways such as SH 86 and the Dharmapuri Main Rd are generally well-maintained. However, caution is advised on smaller bypass roads where potholes or unpaved sections may be encountered. Construction zones near major intersections could pose additional risks.

Alternative Routes

In case of emergencies, consider using the NH 44 for quicker access to emergency services. For bypassing traffic chokepoints, local roads through smaller towns such as Rayakottai and Idappadi may be feasible, although less direct.

Local Regulations on Hazardous Material Transport

Tamil Nadu has specific regulations regarding the transport of hazardous materials, including requirements for safety equipment and adherence to designated routes. Ensure all necessary permits and documentation are in order, and that the vehicle complies with safety standards.

Historical Incidents

Historical data indicates occasional accidents involving heavy vehicles, particularly at high-traffic junctions like Thoppur. There have been instances of overturned vehicles on curves and near urban hubs. Awareness of past incidents can help in preventive planning.

Environmental Considerations

The route passes through environmentally sensitive areas, including forested sections near Dharmapuri. Drivers should exercise caution to avoid spills and littering, with a focus on maintaining the ecological balance.

Communication Coverage

Overall communication is likely reliable due to the presence of major towns and highways. However, rural sections may have inconsistent coverage. Effective communication systems for drivers are crucial for safety and coordination.

Emergency Response Times

Emergency response times can vary widely. In urban zones like Hosur and Salem, responses may be faster (within 30 minutes). In contrast, more remote rural areas could experience delays of up to an hour or more due to transportation and access challenges.

Summary of Risk Assessment

Overall, this route poses a moderate risk level for heavy vehicles carrying hazardous materials. The main concerns are related to weather conditions, road quality, traffic congestion, and communication issues in remote areas. Proper planning, regular communication checks, and adherence to regulations can mitigate these risks significantly. Regular updates on weather and traffic conditions, as well as emergency contact information, should be made available to drivers for a safer journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44077, 77.87533	15 KM/Hr
4	Turn	High	11.44898, 77.87410	15 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45545, 77.81812	30 KM/Hr
8	Turn	High	11.45584, 77.81695	15 KM/Hr
9	Turn	Medium	11.45656, 77.81709	30 KM/Hr
10	Turn	High	11.49433, 77.86696	15 KM/Hr
11	Turn	Medium	11.52596, 77.85527	30 KM/Hr
12	Turn	Medium	11.52797, 77.85580	30 KM/Hr
13	Turn	Medium	11.53825, 77.85096	30 KM/Hr
14	Turn	Medium	11.55713, 77.85823	30 KM/Hr
15	Turn	Medium	11.56416, 77.85968	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	High	11.57854, 77.85429	15 KM/Hr
17	Turn	High	11.58025, 77.85746	15 KM/Hr
18	Turn	Medium	11.58806, 77.85776	30 KM/Hr
19	Turn	Medium	11.60089, 77.84909	30 KM/Hr
20	Turn	Medium	11.60089, 77.84909	30 KM/Hr
21	Turn	High	11.58806, 77.85776	15 KM/Hr
22	Turn	Medium	11.59402, 77.86978	30 KM/Hr
23	Turn	Medium	11.59261, 77.90697	30 KM/Hr
24	Turn	Medium	11.63768, 77.94809	30 KM/Hr
25	Turn	Medium	11.64121, 77.95138	30 KM/Hr
26	Turn	Medium	11.64323, 77.95184	30 KM/Hr
27	Turn	Medium	11.64587, 77.95108	30 KM/Hr
28	Turn	Medium	11.67896, 77.96882	30 KM/Hr
29	Turn	Medium	11.68439, 77.97529	30 KM/Hr
30	Turn	Medium	11.70049, 77.97964	30 KM/Hr
31	Turn	High	11.70179, 77.97900	15 KM/Hr
32	Turn	Medium	11.71394, 77.99572	30 KM/Hr
33	Turn	Medium	11.71739, 78.00032	30 KM/Hr
34	Turn	Medium	11.71984, 78.01216	30 KM/Hr
35	Turn	Medium	11.72770, 78.02297	30 KM/Hr
36	Turn	High	11.73787, 78.03528	15 KM/Hr
37	Turn	High	11.73655, 78.03891	15 KM/Hr
38	Turn	High	11.74257, 78.04708	15 KM/Hr
39	Turn	Medium	11.78092, 78.05792	30 KM/Hr
40	Turn	Medium	11.86790, 78.08347	30 KM/Hr
41	Turn	Medium	11.87902, 78.08209	30 KM/Hr
42	Turn	Medium	11.91357, 78.05868	30 KM/Hr
43	Turn	Medium	11.92556, 78.06499	30 KM/Hr
44	Turn	Medium	11.93424, 78.06331	30 KM/Hr
45	Turn	Medium	11.93805, 78.05273	30 KM/Hr
46	Turn	Medium	11.95639, 78.04760	30 KM/Hr
47	Turn	Medium	11.95796, 78.04956	30 KM/Hr
48	Turn	Medium	11.96062, 78.04957	30 KM/Hr
49	Turn	High	11.96366, 78.05142	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
50	Turn	High	11.96637, 78.04923	15 KM/Hr
51	Turn	Medium	11.96859, 78.05102	30 KM/Hr
52	Turn	Medium	12.12094, 78.13269	30 KM/Hr
53	Turn	Medium	12.28202, 78.06825	30 KM/Hr
54	Turn	Medium	12.50522, 78.03132	30 KM/Hr
55	Blind Spot	Blind Spot	12.51384, 78.03205	10 KM/Hr
56	Turn	Medium	12.51372, 78.03018	30 KM/Hr
57	Turn	Medium	12.51637, 78.02568	30 KM/Hr
58	Turn	Medium	12.52927, 78.02022	30 KM/Hr
59	Turn	Medium	12.67951, 77.86168	30 KM/Hr
60	Turn	Medium	12.68963, 77.85524	30 KM/Hr
61	Turn	Medium	12.69471, 77.85391	30 KM/Hr
62	Turn	High	12.71955, 77.83556	15 KM/Hr
63	Turn	High	12.71967, 77.83554	15 KM/Hr
64	Turn	Medium	12.72339, 77.83973	30 KM/Hr
65	Turn	Medium	12.72557, 77.84045	30 KM/Hr
66	Turn	High	12.72762, 77.84303	15 KM/Hr
67	Turn	Medium	12.72838, 77.84269	30 KM/Hr
68	Turn	Medium	12.74549, 77.81549	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
3	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
4	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
5	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
7	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
8	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
12	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
13	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
14	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
15	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
16	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
17	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
18	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
19	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
20	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
21	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
22	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
26	pharmacy	H2 Pharmacy	12.7301802, 77.8401037	30 km/h	Medium
27	hospital	Surya Multispeciality Hospitals Ind Pvt Ltd	12.7329337, 77.8259093	30 km/h	Medium
28	hospital	Velmath Maternity Home	12.7353452, 77.8261929	30 km/h	Medium
29	hospital	Vijay Hospital, Krishnagiri	12.7350679, 77.825517	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
30	hospital	S.B.S Hospital	12.7338924, 77.8259338	30 km/h	Medium
31	hospital	Uma Kumar Hospital	12.7362537, 77.8243053	30 km/h	Medium
32	hospital	Dr Agarwals Eye Hospital Ltd, Hosur	12.7405888, 77.8205797	30 km/h	Medium
33	hospital	Vasan Eye Care Hospital-Hosur	12.7414997, 77.8187729	30 km/h	Medium
36	hospital	Santhi Hospital & Santhi Jananam Fertility Centre Hosur	12.7488623, 77.8058879	30 km/h	Medium
37	hospital	BMS Multi-speciality Hospital	12.7483681, 77.8063928	30 km/h	Medium
38	hospital	Janani Hospital	12.7520976, 77.8031203	30 km/h	Medium
39	hospital	ESI Krishnagiri	12.7570375, 77.7990046	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	Government Higher Secondary School	12.5844162, 77.9669764	30 km/h	Medium
24	school	Panchayat Union Primary School	12.6078768, 77.9381865	30 km/h	Medium
25	school	Panchayat Union Primary School	12.6835154, 77.8604798	30 km/h	Medium
34	school	St. Joseph's Primary School	12.7451275, 77.8131685	30 km/h	Medium
35	school	St. Joseph's Matriculation Higher Secondary School	12.7463204, 77.8072784	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44077, 77.87533



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49433, 77.86696



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52596, 77.85527



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52797, 77.85580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53825, 77.85096



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55713, 77.85823



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56416, 77.85968



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57854, 77.85429



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58025, 77.85746



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58806, 77.85776



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60089, 77.84909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60089, 77.84909



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58806, 77.85776



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59402, 77.86978



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63768, 77.94809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64121, 77.95138



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64323, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64587, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67896, 77.96882



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.70049, 77.97964



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70179, 77.97900



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71394, 77.99572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71739, 78.00032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71984, 78.01216



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72770, 78.02297



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73655, 78.03891



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.74257, 78.04708



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78092, 78.05792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86790, 78.08347



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87902, 78.08209



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91357, 78.05868



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92556, 78.06499



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93424, 78.06331



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93805, 78.05273



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95639, 78.04760



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95796, 78.04956



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96062, 78.04957



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96366, 78.05142



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96637, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96859, 78.05102



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50522, 78.03132



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51384, 78.03205



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51372, 78.03018



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.51637, 78.02568



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52927, 78.02022



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.67951, 77.86168



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.68963, 77.85524



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.69471, 77.85391



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71955, 77.83556



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.71967, 77.83554



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72339, 77.83973



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.72557, 77.84045



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.72762, 77.84303



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.72838, 77.84269



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.74549, 77.81549
